

Anomaly Detection Autoencoder - High-Level Project Steps

Flowchart summary of the notebook from PaySim transactions to suspicious transaction detection

1. Load PaySim transaction data

Load financial transaction records containing transaction type, amount, balances, fraud label, and rule-based fraud flag.



2. Create practical working sample

Keep a manageable normal sample and all fraud transactions for evaluation so local training remains practical.



3. Select model features

Use transaction behaviour features and exclude isFraud, isFlaggedFraud, and account ID columns to avoid leakage.



4. Preprocess features

Scale numeric features and one-hot encode transaction type using a fitted training preprocessor.



5. Split train, validation, and test

Train and validate only on normal transactions; test on both normal and fraud transactions.



6. Train dense autoencoder

Train the autoencoder to reconstruct normal transaction feature vectors. Input and target are the same.



7. Calculate reconstruction error

Generate one reconstruction error score for each test transaction.



8. Set anomaly threshold

Use the 95th percentile of normal validation error as the suspicious transaction cutoff.



9. Flag suspicious transactions

Mark transactions above the threshold as suspicious and compare against fraud labels.



10. Evaluate and tell the BI story

Use precision, recall, F1-score, threshold trade-off, confusion matrix, and KPI charts for business interpretation.

Final business outcome

The dense autoencoder learns normal transaction behaviour and creates reconstruction-error anomaly scores. Fraud labels are not used during training; they are used only later to evaluate and explain the suspicious transaction signal.